anticipated were welcomed. For instance, the theory of canonizers (Section 6.3) was not initially known to the first project participants, but was brought to the attention of the project by a later contributor. Conversely, while there were hopes expressed early in the project that modern large language models (LLMs) could automatically generate many of the proofs required, it turned out in practice that other forms of automation, particularly ATPs, were significantly more effective at this task (at least if one restricted to publicly available LLMs), and the project largely moved away from the use of such LLMs (other than to help create the code for the visualization tools).

There are several mathematical and computational questions that could potentially be addressed in future work building upon the outcomes of ETP. Here is a list of some possible such future directions.

(1) Does the law $\mathrm{x} \simeq \mathrm{y} \diamond (\mathrm{x} \diamond ((\mathrm{y} \diamond \mathrm{x}) \diamond \mathrm{y}))$ (E677) imply $\mathrm{x} \simeq ((\mathrm{x} \diamond \mathrm{x}) \diamond \mathrm{x}) \diamond \mathrm{x}$ (E255) for finite magmas, i.e., E677 $\models_{\mathrm{fin}}$ E255? This is the last remaining implication (up to duality) for finite magmas to be resolved. A number of partial results on this problem may be found at `https://teorth.github.io/equational_theories/blueprint/677-chapter.html`.
(2) The ETP focused on determining relations $\mathrm{E} \models \mathrm{E}'$ between one law and another. Could the same methods also systematically determine more complex logical relations, such as $\mathrm{E}_1 \wedge \mathrm{E}_2 \models \mathrm{E}_3$, for all laws $\mathrm{E}_1, \mathrm{E}_2, \mathrm{E}_3$ in a specified set? This includes the question of implications between equational laws in semigroups (associative magmas). One could also consider implications involving magma properties that are not equational laws, such as cancellability or existence of a unit element.
(3) Call an implication $\mathrm{E}_1 \models \mathrm{E}_2$ "irreducible" if there is no equational law E with $\mathrm{E}_1 \models \mathrm{E} \models \mathrm{E}_2$, other than those laws equivalent to either E_1 or E_2. For instance, E2 $\models$ E4 is irreducible, since E4 implies any law of the form $w \simeq w'$ where the left-most variable of w matches the left-most variable of w'. On the other hand, E4 in conjunction with any law not of that form yields E2. Similar *ad hoc* arguments can produce other irreducible implications, e.g., E2 $\models$ En for $n = 5, 895, 26302$. Could one replicate the ETP to classify all stable implications among the same 4694 equations studied in this project?
(4) For a given finite non-implication $\mathrm{E}_1 \not\models_{\mathrm{fin}} \mathrm{E}_2$, are there bounds on the proportion of variable assignments for which E_2 holds, similarly to how in a finite group either all elements square to the neutral element, or at most 3/4 of them do?

Some other directions do not concern implications between laws, but may benefit from data generated by the ETP.

(5) Does the law $\mathrm{x} \simeq \mathrm{y} \diamond (\mathrm{y} \diamond (\mathrm{y} \diamond (\mathrm{x} \diamond (\mathrm{z} \diamond \mathrm{y}))))$ (E5093) have any infinite models? In [34] it was shown that it has no non-trivial finite models, but the infinite model case was left as an open question. A partial classification of laws of order 5 with infinite models but no finite models is given at `https://teorth.github.io/equational_theories/blueprint/order-5-austin-laws.html`.
(6) A key feature of finite magmas $\mathcal{M}$ is that they are surjunctive, in the sense that any definable map from M to itself that is injective, is also surjective (or vice versa), where "definable" is with respect to the language of magmas. Are there equational

theories that admit surjunctive models, but yet do not have any non-trivial finite models?

(7) Are all finite weak central groupoids, namely magmas obeying $x \simeq (y \diamond x) \diamond (x \diamond (z \diamond y))$ (E1485), necessarily of size n^2 or $2n^2$? More generally, what is the spectrum of each law or conjunction of laws, and what are the possible asymptotics for the fine spectrum in terms of model size?

(8) How "stable" is a given law E? For instance, if a finite magma satisfies a law E some proportion $1 - \varepsilon$ of the time, with ε small, can the magma be perturbed into one that satisfies E exactly? Related to this is the question of whether a law E is "rigid" or "mutable": is it possible to add an element or to make a small number of modifications to a magma satisfying E, in a way that still preserves E? Such properties helped suggest whether certain magma construction techniques, such as modifying a base magma, were likely to be successful.

(9) For each law, can its free magma with one or more generators be described explicitly?

(10) Which laws admit an interesting theory of smooth magmas, analogous to Lie groups?

14.1. **Miscellaneous remarks.** It is possible that the timing in which certain proof methods were introduced into the project created some opportunity costs. For instance, by deploying automated theorem provers at an early stage, we might have settled some implications that had more interesting human-readable proofs that we missed. Similarly, we developed some sophisticated theory for the equation E854, such as Corollary 6.12, that is now superseded by finite counterexamples; but had the finite counterexamples been discovered first, we would not have found the theoretical arguments. It may be productive for future work to revisit some portions of the implication graph and locate alternate proofs and methods.

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Appendix A. Numbering system

In this section we record the numbering conventions we use for equational laws.

For this formal definition we use the natural numbers $0, 1, 2, \ldots$ to represent and order indeterminate variables; however, in the main text, we use the symbols $\mathrm{x}, \mathrm{y}, \mathrm{z}, \mathrm{w}, \mathrm{u}, \mathrm{v}, \mathrm{r}, \mathrm{s}, \mathrm{t}$ instead (and do not consider any laws with more than eight variables).

To define the ordering we use on equational laws, we first consider the case where there is a single indeterminate $*$. We place a well-ordering on words w, w' with a single indeterminate $*$ by declaring $w > w'$ if one of the following holds:

- w has a larger order than w'.
- $w = w_1 \diamond w_2$ and $w' = w'_1 \diamond w'_2$ have the same order $n \geq 1$ with $w_1 > w'_1$.
- $w = w_1 \diamond w_2$ and $w' = w'_1 \diamond w'_2$ have the same order $n \geq 1$ with $w_1 = w'_1$ and $w_2 > w'_2$.

Thus

$$\begin{aligned} * < * \diamond * < * \diamond (* \diamond *) < (* \diamond *) \diamond * \\ < * \diamond (* \diamond (* \diamond *)) < * \diamond ((* \diamond *) \diamond *) < \ldots \end{aligned}$$

We similarly place a well-ordering on equational laws $w_1 \simeq w_2$ with a single indeterminate $*$ by declaring $w_1 \simeq w_2 > w'_1 \simeq w'_2$ if one of the following holds:

- $w_1 \simeq w_2$ has a larger order than $w'_1 \simeq w'_2$.
- If $w_1 \simeq w_2$ has the same order as $w'_1 \simeq w'_2$, and $w_1 > w'_1$.
- If $w_1 \simeq w_2$ has the same order as $w'_1 \simeq w'_2$, $w_1 = w'_1$, and $w_2 > w'_2$.

Thus for instance

$$(* \diamond * \simeq * \diamond (* \diamond *)) < (* \diamond * \simeq (* \diamond *) \diamond *).$$

Finally for equational laws with alphabet $\mathrm{x}, \mathrm{y}, \mathrm{z}, \mathrm{w}, \mathrm{u}, \mathrm{v}, \mathrm{r}, \mathrm{s}, \mathrm{t}$, define the *shape* of that law to be the law formed by replacing all indeterminates with $*$; for instance, the shape of $\mathrm{x} \diamond (\mathrm{y} \diamond \mathrm{z}) = (\mathrm{x} \diamond \mathrm{y}) \diamond \mathrm{z}$ (E4512), is $* \diamond (* \diamond *) \simeq (* \diamond *) \diamond *$. We then place a well-ordering $w_1 \simeq w_2$ with indeterminates $\mathrm{x}, \mathrm{y}, \mathrm{z}, \mathrm{w}, \mathrm{u}, \mathrm{v}, \mathrm{r}, \mathrm{s}, \mathrm{t}$ by declaring $w_1 \simeq w_2 > w'_1 \simeq w'_2$ if one of the following holds:

- The shape of $w_1 \simeq w_2$ is greater than the shape of $w'_1 \simeq w'_2$.
- $w_1 \simeq w_2$ and $w'_1 \simeq w'_2$ have the same shape, and the string of variables appearing in $w_1 \simeq w_2$ is lower in the lexicographical ordering (using $\mathrm{x} < \mathrm{y} < \mathrm{z} < \mathrm{w} < \mathrm{u} < \mathrm{v} < \mathrm{r} < \mathrm{s} < \mathrm{t}$) than the corresponding string for $w'_1 \simeq w'_2$.

Thus for instance any law of shape $* \diamond * \simeq * \diamond (* \diamond *)$ is lower than any law of shape $* \diamond * \simeq (* \diamond *) \diamond *$. Among the laws of shape $* \diamond * \simeq * \diamond (* \diamond *)$, the lowest is $\mathrm{x} \diamond \mathrm{x} \simeq \mathrm{x} \diamond (\mathrm{x} \diamond \mathrm{x})$, which is less than (say) $\mathrm{x} \diamond \mathrm{x} \simeq \mathrm{y} \diamond (\mathrm{y} \diamond \mathrm{y})$, which is in turn less than $\mathrm{x} \diamond \mathrm{y} \simeq \mathrm{x} \diamond (\mathrm{x} \diamond \mathrm{x})$.

We say that two equational laws are *definitionally equivalent*[40] if one can be obtained from another by some combination of relabeling the variables and applying the symmetric law $w_1 \simeq w_2 \iff w_2 \simeq w_1$. For instance, $(0 \diamond 1) \diamond 2 \simeq 1$ is definitionally equivalent to

[40]This can be distinguished from the weaker notion of *propositional equivalence* (mutual entailment) used in the rest of the paper.